Completion of the ambient paired-head branch

Human Compression project: L044 continuation

Retain the complete standing hypotheses and smaller-degree induction of *Critical odd-degree tails behind exceptional ends*. Thus $G \le A_n$ has a connected string C-marking, $n \ge 12$, and every maximal ambient omission is intransitive. Eligibility below always refers to that ambient omission, not to a restricted tail.

Theorem 1. *In either of the following source-head configurations, suppose that each colour $j \ge k$ has two ambient-eligible edges within some single actual $H = G_{\ge k}$-orbit:*

1. *G_0 has the two orbits $\{a\}$ and its complement; $k = 2$ and $b = a^{\rho_0}$.*
2. *The first three generators are the actual supported six-point head from source Table 2(2) or (3), in an extendable orientation, with $k = 5$. The accepted exceptional-head interface is retained: $D = \{1, \dots, 6\}$, $\tau = \rho_3 = (6\ b)\delta$, H fixes $D \cup \{b\}$, and $G_{\ge 4}$ is transitive on $\Omega \setminus D$.*

Then $r \le \lfloor (n-1)/2 \rfloor$.

Proof. Suppose the rank bound fails. The complete critical-head reduction gives $n = 2r$, with the actual natural alternating tail used by the exceptional-head proofs. In the singleton case $n \ge 14$, and $H = G_{\ge 2}$ fixes a, b and is intransitive on its support. In the exceptional case $n \ge 18$ and H fixes its seven-point head.

Choose one such orbit and pair for each tail colour. The whole-orbit transport theorem normalizes this assignment so that its full local $K = G_{>k}$ is intransitive in each assigned orbit. Every colour is assigned exactly once; the assigned colour sets are disjoint. For the actual nontrivial H-orbits O_s put $d_s = |F_s| - |O_s|/2$, and let $f = |\operatorname{Fix}(H)|$. The normalized bounds give all $d_s \le 0$ and

$$S = \sum_s d_s = f/2 - k \ge -3/2. \tag{1}$$

Let $\tau = \rho_0$ in the singleton case and $\tau = \rho_3$ in the exceptional case. This involution centralizes H and belongs to every ambient omission of a tail colour. On the specified fixed head it is the single transposition $(a\ b)$ or $(6\ b)$, respectively. Since τ is even, its action on the complement of that head is odd.

Centralization permutes actual H-orbits equivariantly. An exchanged pair of nontrivial orbits contributes an odd sign only when their common size is odd; the accepted equivariant aggregation theorem then gives $S \le -3$, contrary to (1). On a preserved nontrivial orbit, a nonidentity centralizing involution is free. Its sign is odd only on an orbit of size $2q$ with q odd. If the assignment is nonempty, the complete strict-internal and localized-internal proofs exclude it in both heads, including the final first-assigned odd-anchor configuration. If the assignment is empty and $q \ge 3$, its exact deficit is $-q \le -3$, again contrary to (1). The case $q = 1$ is the empty internal pair excluded by the localized two-point theorem.

It remains to include the H-fixed points, which are absent from its support. The full ambient interval theorem gives

$$2(r-k) \le n - f - c, \qquad f + c \le 2k, \tag{2}$$

where $c \ge 1$ counts the nontrivial H-orbits. In the singleton case $f \ge 2$ and $c \ge 2$ force $f = c = 2$; there are no additional fixed points outside $\{a, b\}$. In the exceptional case, if the action on the

additional fixed points is odd, it exchanges at least two of them. The complete localized two-point theorem excludes exactly this possibility; its reduction from (2) gives $f = 9, c = 1$ before applying the actual pure-three-cycle/IP obstruction.

These cases exhaust every odd contribution to τ outside its head: preserved nontrivial orbits, exchanged nontrivial orbits, and exchanged singleton orbits. Each is impossible. This contradicts the required odd sign, proving the bound. □

Corollary 2 (The precise residual source interface)**.** *Under either critical head and rank failure, some tail colour has at most one ambient-eligible edge in every actual H-orbit.*

Proof. Otherwise choose an orbit with two eligible edges independently for each colour and apply the theorem. Such a choice does not require different colours to use different orbits. □

Attribution and limits. This combines the complete paid orbit-transfer, equivariant-deficit, strict-internal and localized-interval proofs for the all-paired part of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.31, physical pages 42–44. It uses the source's deficit strategy, with the stated replacement proofs and an explicit accounting of singleton orbits. No stronger literal local-deficit or diagram assertion is inferred from the final contradiction.

The condition is that each colour has an eligible pair *in one actual orbit*. Two edges split across different orbits do not satisfy it. Nor does a pair crossing only a local tail omission. The corollary identifies, rather than resolves, that remaining source-routing branch. Complete source recovery, rigidity/table contracts, formalization and original or question credit remain outside this note.